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THE ROLE OF TRANSVERSE AIR JETS IN THE FORMATION OF GAS TURBINE EMISSIONS

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Abstract

Results are presented of emissions and lean blowout measurements in a generic combustor primary zone for the effects of opposed, circular combustion air jets of a range of sizes, positioned half a dome height downstream from the dome. The primary zone was operated at simulated engine high power conditions, and representative dome and liner pressure drops. It is shown that most of the NOx is generated in the jets. Although an optimum jet size is identified, the lowest overall emissions resulted from a no combustion air jet configuration; air jets are necessary however, to confer lean blowout stability and short flame length.

Introduction

In 1987 the U.S. Air Force Wright Laboratory, Aero Propulsion and Power Directorate (WL/PO), Wright-Patterson Air Force Base, Ohio, established the *Combustor Design Model Evaluation (CDME) Program.* This effort is a comprehensive joint Government, Industry and University research program aimed at improving the design and analysis capabilities for the combustors of aircraft gas turbine engines. The CDME program is addressing many important aspects of combustor design.

One study that is currently in progress is an investigation into the minimization of engine exhaust emissions, and the relationships between generation of these emissions and combustor operability characteristics. As part of this study a broad-ranging investigation of combustor primary zone behavior is being conducted.

Program Objectives

The overall intent of this part of the CDME program is to understand the detailed behavior of the major flame holding of a modern generic combustor for aircraft gas turbine engines, and to establish an appropriate database suitable for design purposes.

Test Vehicle

A relatively simple combustor developed under an earlier program[1] has been adapted for the experimental study of the influence of simulated engine primary zone flow structures on the emissions generation and lean stability characteristics.

The combustor consists of a single representative fuel injector and its associated dome interface, mounted in a plain bulkhead. The injector is a high-swirl airblast atomizer[2], and the dome interface consists of a small, discrete-jet swirler co-annular with the injector, together with a dome heat shield fed with cooling air that is discharged radially across the bulkhead face. The cross-section of the combustor is square for optical access by laser diagnostics, but generous fillets are provided in the corners to suppress vorticity generation that might result in spurious flameholding. The hydraulic diameter is 153 mm and the length is 490 mm. The combustor outlet is restricted by a 45 percent geometric blockage orifice plate to provide representative back pressure of the unsimulated portion of a real combustor.

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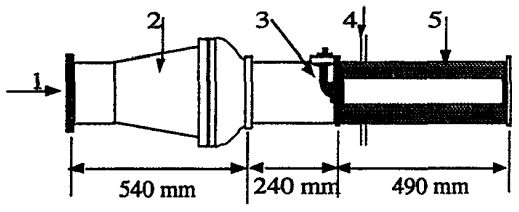
The combustor is mounted horizontally, and the top and bottom walls are removable, uncooled metal plates that accommodate appropriate arrays of combustion air jets. One combustor side-plate (at 90 degrees to the liner plates containing the air jets) contains a fitting for application of a torch ignitor; the other side contains a full-length quartz window for visual access.

Figure 1 shows the combustor with its air entrance conditioning unit. Air supplies to the dome and the combustion air jets are separately metered, and may be electrically preheated to a maximum temperature of about 500F (533K).



1. Air entrance 2. Diffuser 3. Fuel injector
4. Liner air-jets 5. Quartz window

Figure 1a): Laboratory-Scale Gas Turbine Combustor

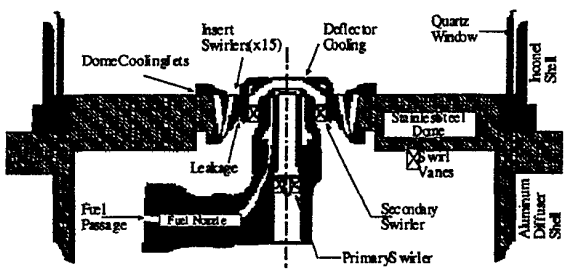


Figure 1b): Combustor Fuel Injector and Dome

Engine Power Simulation

The efficacy of the combustor for its intended purpose is discussed in detail in Reference 2. The combustor represents a "single-cup" planar segment of an annular combustor, and simulates the primary zone only.

It is operated at atmospheric pressure for convenience and cost reasons. Engine high power conditions are simulated by closely matching Spalding's partial modeling flow criterion K_f, reasonably approximating the temperature rise term $\Delta T / T_{in}$, and by substituting gaseous propane for liquid JP/Jet fuel. Together, these ensure that engine flow patterns are closely approximated and that liquid evaporation is not a problem. It was concluded that the combustor can reproduce reasonably well and to an acceptable degree the flow patterns of an engine combustor.

Description of the Rig

The test combustor is installed horizontally in a test cell at Wright Patterson Air Force Base, Ohio. Reference 2 gives a detailed description of the rig and facility in which the combustor is installed. Of relevance here is the fact that inlet air was dried to very low dew points prior to entering the compressors.

Experimental Variables

The array of combustion air ports presently to be described is confined to an opposed pair of circular flush holes (in upper and lower liners) placed inline with the fuel injector centerline. Diameters of 0.25 in. (6.35 mm), 0.375 in. (9.53 mm), 0.50 in. (12.70 mm) and 0.75 in. (19.05 mm) are reported . Although these pairs of jets could be situated 1.0, 0.75 and 0.5 dome heights (H_d) downstream from the dome, only the latter position is used in the results to be presented.

The inlet air temperature was held constant at a nominal 500F (533K). This value was selected in order to first, satisfy the simulation criteria, and second, to ensure the generation of sufficient NOx without a large burnup of CO; thus, trends and effects in these two pollutants could be detected easily and measured accurately.

Pressure drop across a combustor is in practice, not a free variable but is set by engine performance considerations. Generally, the allowable pressure drop across a dome is a little higher than that across liners due to the recovery of some dynamic head from the compressor. For the majority of the work to be presented the dome pressure drop was held fixed at a nominal value of 4.3 percent of the upstream pressure, and the liner pressure drop was 3.0 percent. These values are representative of typical engine conditions. Under these conditions all of the transverse jets with different initial diameters penetrated to the combustor centerline.

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To extend the generality of the results the normal range of liner air pressure drops for each hole size was 1-5 percent of the upstream pressure. To explore the effects of jet turbulence this range was extended to as high as 12 percent for the 0.5 in. (12.70 mm) hole.

Emissions Measurements

The emissions sampling probe is a water-cooled, quick-quench, single-point type, and is made of stainless steel with an Inconel tip. The sample enters a converging nozzle of orifice diameter 0.059 in. (1.5 mm), which then diverges to the 0.180 in. (4.6 mm) I.D. of the stainless tube. This arrangement results in a supersonic expansion which reduces the gas stream temperature. Together with the water cooling of the probe, this achieves quenching of chemical reactions.

The sample is transferred from the probe by heated lines that avoid condensation, to the emissions measuring equipment, which is fairly standard. Again, detailed descriptions are given in Reference 2.

The axial measurement plane for gaseous emissions was 14 in. (356 mm) downstream from the face of the dome. Kinetics calculations indicated that at this position the bulk NOx was not changing too rapidly, and CO consumption rates were reasonably slow. Furthermore, at this axial station the profiles of measured emissions were flat enough over the entire range of equivalence ratios (for all jet sizes at the $H_d/2$ position) for centerpoint and area-weighted mean values to be in exceptionally close agreement, thus permitting centerpoint values to be used for economy, (errors less than ± 10 percent). Since the emissions probe was introduced into the combustor from the rear through the exit orifice plate, the extent of the radial traverses was determined by the size of this orifice plate. The extent of the radial profile traverses was $\pm$ 1.75 in. ($\pm$ 44 mm) from the combustor centerline. Traverses were made both horizontally and vertically.

Comparisons were made of the metered fuel/air ratios to the measured exhaust fuel/air ratios based on the carbon balance. The agreements were better than 10 percent for metered equivalence ratios of 1.0 and less.

As the inlet air was dried in the facility prior to entering the rig the NOx levels to be quoted are all dry values.

Test Procedure

For each air-port size, axial position and pressure drop, the fuel flow rate was varied to cover a range of overall equivalence ratios such that the anticipated minimum in the CO-curve and the maximum in the NOx-curve could be fully established.

It is usual to present emissions measurements as a function of an appropriate equivalence ratio. The presumption is that the flow patterns in the combustor are established by the geometry and remain unchanged with this method (varying fuel flow rate) of operating, thus making equivalence ratio a good single descriptor of the emissions. Since the combustion air jets established by the different liner hole sizes all penetrated to the combustor centerline, this is a reasonable supposition. However, the high engine power simulation assumed that flow patterns were not affected to first order by the fuel flow. With a gaseous fuel having flow rate as a prime operational variable, this assumption must be verified due to fuel momentum considerations.

It is difficult to establish a true uniqueness test of the data due to the large number of variables involved. However, at fixed dome and liner pressure drops turbulence generated will be fixed, and variations in NOx with overall equivalence ratio for fixed jet size can be compared against variations in NOx with overall equivalence ratio for fixed fuel flow rate but with a range of combustion air jet initial diameters, i.e. variable airflow.

Such a comparison is complicated by the postulated mechanism for jet NOx formation[2] (see below also), which is proportional to jet size. However, provided that jet NOx is not a large proportion of the total NOx (small jets and/or low equivalence ratios), and, provided that the comparison is restricted to jet sizes equal to or less than the fixed jet size of the comparison, and that the range of constant fuel flows examined is selected to be within the range of flows for the fixed jet data, such comparisons may be made.

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Figure 2 compares NOx emission indices (EI's in gm NOx as NO_2/kg fuel burned) data for the 0.5 in. (12.7 mm) jet case over a range of equivalence ratios obtained by varying fuel flow, with NOx data for 0.5 in. (12.7 mm), 0.375 in. (9.5 mm) and 0.25 in. (6.35 mm) at fixed fuel flow rates of 0.075, 0.100, 0.125 and 0.150 lbm/min. (0.034, 0.045, 0.057 and 0.068 kg/min. respectively). Rich and lean points are included.

The plotted data in Figure 2 on both lean and rich sides correlate within the measurement accuracy and repeatability of the single-test data scatter[2].

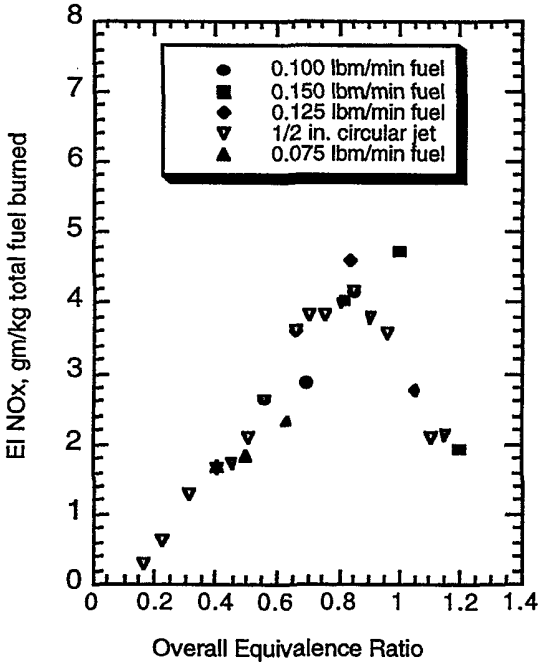

Figure 2: Uniqueness Plot Based on NOx, to Evaluate Fuel Momentum Effects

The conclusion can therefore be drawn that fuel momentum does not affect the flow field, and as a consequence, that for fixed pressure drop, equivalence ratio is a unique parameter describing NOx.

Dome Characteristics

The combustor can be operated with zero combustion air jets to obtain the characteristics of the dome alone.

Figure 3 is a plot of the emission index (EI) of NOx for the dome alone, as a function of dome equivalence ratio (metered). As expected, the measured NOx reaches a peak of about 3.6 at a dome equivalence ratio of 0.82. As the dome equivalence ratio exceeds unity NOx falls due to decreasing bulk flame temperatures, but unburned fuel leaves the combustor.

Also shown on Figure 3 are calculated NOx curves using two approaches. First, assuming "experimentally-perfect premixing" of fuel and air, and second, representing the dome flow field as a reactor network having a degree of partial stirring.. For the "experimentally-perfect premixing" case the empirical correlation of Roffe & Venkataramani[3] for lean propane and air systems was used. In this correlation the residence times were estimated for volumes based on observed flame dimensions. The reactor network was based on computational fluid dynamic calculations for this injector and hole pattern[1], with reactor dimensions based on observed flame dimensions and having fuel heating value reduced to account for estimated heat losses from the combustor. The network consisted of a well-stirred reactor with a parallel plug-flow reactor that processed 13 percent of the total flow that was recirculated back into the dome.

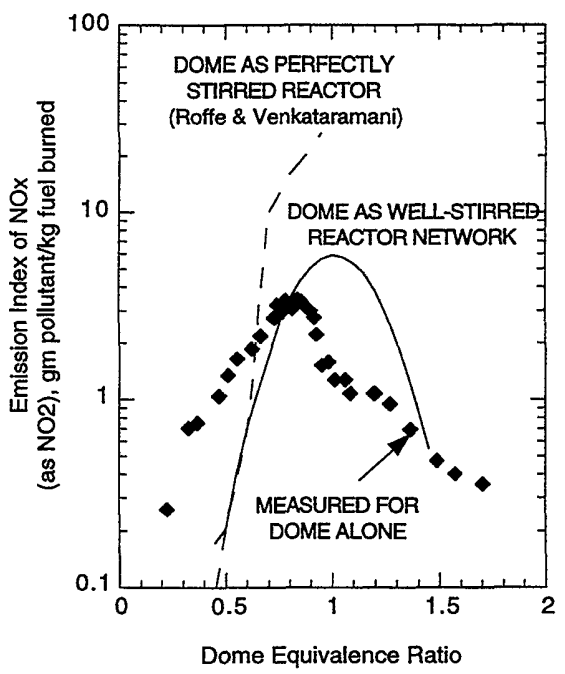

Figure 3: Assessment of Dome NOx Characteristics

The effect of improved mixing as represented in the two calculations, is to increase the peak NOx and to narrow the NOx curve across equivalence ratio; furthermore, the peak NOx occurs at dome equivalence ratios around unity.

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Comparing the measured dome NOx against the two calculations demonstrates that despite the use of a gaseous fuel and the intimate association of fuel and air in the fuel injector, the fuel is burned in the dome in far from premixed form, and consequently, exhibits some degree of diffusion flame burning.

Overall Burning Characteristics

In Reference 2 the NOx data were analyzed in terms of a rich/lean burning system, where excess fuel from a fuel-rich dome was assumed to burn fuel-lean in the combustion air jet system, as was described in Reference 4. If this is so, the question concerning the nature of the flame in these two burning zones arises. It was assumed in Reference 2 that it was the same in both zones, i.e. based on the previous section, flames exhibiting some degree of diffusion burning.

Figure 4 is a plot of overall NOx EI's for several jet systems at $H_d/2$ from the dome, including the dome alone, plotted against dome equivalence ratio, at fixed pressure drops.

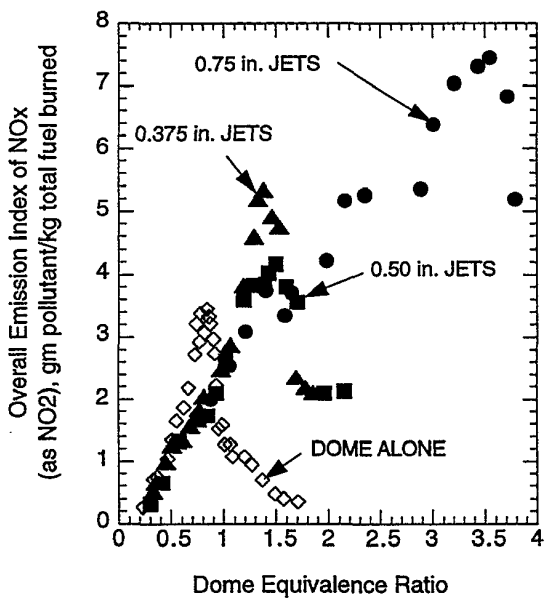

Figure 4: Overall Burning Characteristics for NOx

The characteristic of the dome-alone NOx is that from Figure 3. The characteristics for the individual combustion air jet cases collapse onto a common curve for dome equivalence ratios up to about 1.15. For dome equivalence ratios of 0.55 and less, this collapsed data is co-incident with the dome alone data. This behavior indicates that basic flame characteristics in the dome are unaffected by the presence of the jets.

The NOx peak for jets being present is shifted to higher dome equivalence ratios, in the range 1.3 - 1.55, from the nominal 0.82 equivalence ratio for the dome alone. The levels of NOx at the peak, and the subsequent behavior on the dome rich-side of the peak, are dependent on the individual jet size.

The shifted NOx peak from 0.82 with no jets to 1.3 - 1.55 dome equivalence ratio with jets, indicates that in addition to peak temperatures being reached at lower equivalence ratios (0.82 rather than about 0.92-1.0)) due to unmixedness effects, some of the combustion jet air is actually active in the dome. The co-incidence of the data for individual jets at equivalence ratios less than 1.15 suggests that the dome aerodynamics are not strongly influenced, or least modified, by the details of the air jets, i.e. that any additional air introduced into the dome takes part in chemical reactions in a region somewhat separate from the jets themselves, and, that this region is not far upstream in the dome itself. Flame photographs identify a small luminous region on the injector centerline, just upstream of the confluence of the transverse combustion air jets. CFD analysis of this type of combustion system[1], identifies a well-mixed recirculation bubble in this position. This bubble could be where the small quantity of jet air is utilized in the dome. The centerline penetration of all jets at 3 percent liner pressure drop ensures that this bubble is present whenever the jets are present.

It will be noted that for the 0.75 in. (19.05 mm) jets, where, as was shown in Reference 2, most of the total NOx produced originates in the jets, high levels of NOx are eventually produced after the initial NOx-peak that denotes a stoichiometric dome. This is mainly due to jet NOx as the equivalence ratio increases. These high levels of NOx fall onto the same general curve as for the other jets, and for the dome alone at dome equivalence ratios less than 0.55. The existence of this "common curve" indicates that that there is no substantial difference in the flame characteristics of dome and jets, and, that the NOx produced in both regions is produced by the same kinetic mechanism.

Jet System Behavior

In Reference 2 the adopted analysis strategy for emissions data reduction was one of

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treating the generic primary zone as being a quasi-staging[4], rich/lean burning combustor. The dome region was generally fuel-rich, and the importance of NOx generation in the "lean burning" combustion air jet system itself was highlighted.

This aspect is illustrated here in Figure 5 where the percentage of total NOx that is generated in the jets for each initial jet size is plotted in terms of overall equivalence ratio. The jet NOx was obtained from the total NOx at given overall equivalence ratio by subtracting from it the dome-alone NOx at an appropriate dome equivalence ratio. Such a procedure is only valid if the predominant mechanism for NOx generation is the same in both regions of the combustor. As was demonstrated above, this appears to be the case. The percent NOx was obtained by multiplying the calculated emission indices for the jets by the mass flow rate of fuel burned in the jets, and dividing by the measured overall emission index for the primary zone at the same equivalence ratio, multiplied by the total fuel flow rate.

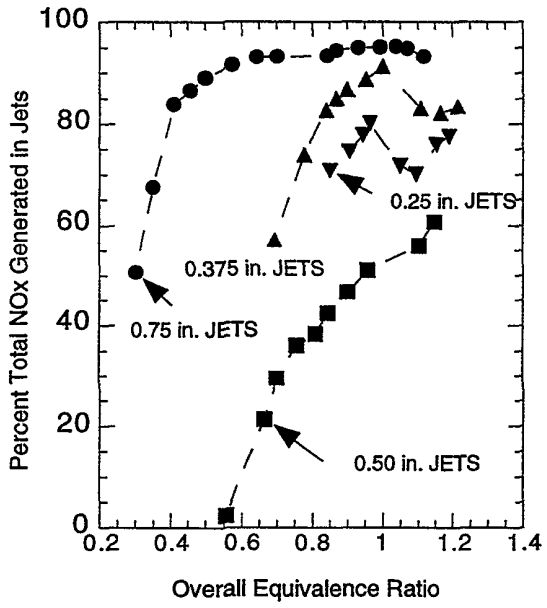

Figure 5: Contribution of Jet-Generated NOx

Figure 5 shows that for the 0.75 in. (19.05 mm) initial jet size more than half of the NOx originates in the jets for all operating equivalence ratios; for the 0.50 in. (12.7 mm) jet the jet NOx only reaches 50 percent of the total for a stoichiometric primary zone. The general behavior is for the percent jet NOx to increase with overall equivalence ratio, and to do so rather steeply initially until a plateau level is reached. This behavior is repeated for each jet system, and the plateau level is lower for smaller jets than for the larger jets. For the 0.75 in. (19.05 mm) jets the plateau is for about 95 percent jet NOx; for the 0.375 in. (9.53 mm) jets it is about 85 percent, and for the 0.25 in. (6.35 mm) jets it is about 75 percent. The 0.5 in. (12.70 mm) jet system has not yet reached its plateau level in the range of overall equivalence ratios covered. This jet system is the exception to the general behavior. If overall equivalence ratio is increased above unity, percent combustion air jet NOx contribution to overall NOx would eventually start to fall as the jet system became over-rich. This is clearly evident for the two smaller jets.

The 50 percent jet NOx condition is reached at overall equivalence ratios that increase with decreasing jet size, and that corresponds to the condition where the dome equivalence ratio exceeds unity in each case. For the 0.50 in. (12.70 mm) jet system the 50 percent jet NOx condition is reached at an overall equivalence ratio of 0.96. From stoichiometry plots this corresponds to a dome equivalence ratio of 1.7. This jet system generates lower jet NOx levels at all overall equivalence ratios than either larger or smaller jets. Clearly, the 0.5 in. (12.70 mm) jet system behaves differently to all of the others.

Figure 6 is a plot from Reference 2 with additional information added for the smaller jets. The plot shows the EI's of jet NOx for each jet as a function of jet equivalence ratio.

For jet equivalence ratios less than 0.8-0.9 (where a NOx peak is expected) the NOx for the 0.375 (9.53 mm), 0.50 in. (12.7 mm) and 0.75 in. (19.05 mm) jets is essentially independent of jet equivalence ratio. This suggests that the fuel burned in the jets at these lower equivalence ratios is reacted in stoichiometric interfaces associated with the fresh air entering through the jets. The jet NOx EI's consistently begin to decrease at higher jet equivalence ratios. This is because unburned fuel is now exiting the combustor. The jet NOx EI plateau levels for the lower jet equivalence ratios decrease with increasing jet initial diameter, with the exception of the 0.50 in. (12.7 mm) jet system, which again has the lowest values. For the 0.25 in. (6.35 mm) jets the jet NOx EI's continuously decrease with increasing jet equivalence ratio over the entire range covered. This suggests that there is something different about this jet system.

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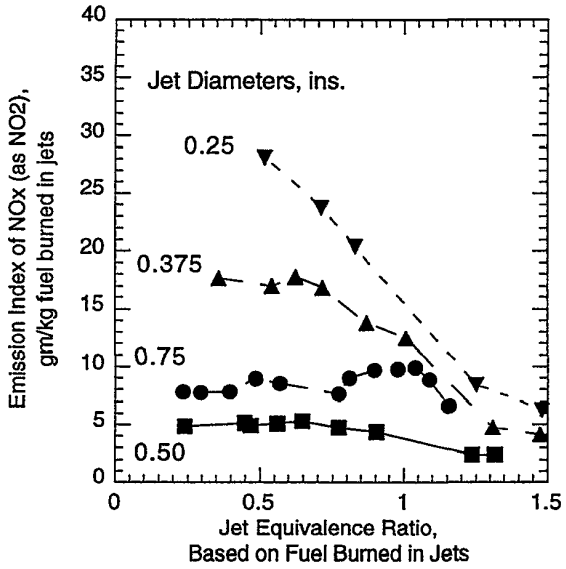

Figure 6: Demonstration of Stoichiometric Burning in Jet Systems

When the absolute jet NOx, in gm/min., is plotted against jet equivalence ratio, Figure 7, the behavior exhibits the characteristic equivalence ratio dependency, with NOx peaks occurring in the jet equivalence ratio range of 0.8-1.0. The absolute NOx levels decrease with decreasing jet size, which is the opposite behavior to that of the EI's. This is accounted for by the different amounts of the total fuel actually burned in the jets, where the 0.75 in. (19.05 mm) jet burns the most fuel and the 0.25 in. (6.36 mm) the least, and where, for a given jet size, the amount of jet fuel decreases with decreasing overall equivalence ratio. Once again, the 0.50 in. (12.7 mm) jet system shows as the optimum, having the lowest absolute jet NOx.

The optimum behavior of the 0.50 in. (12.7 mm) jet system is demonstrated in Figure 8 where the EI's of jet NOx are plotted against jet diameter for three primary zone (overall) equivalence ratios in the range of peak bulk-flame temperatures. Note that for all jet sizes, the jet NOx EI's decrease with increasing primary zone equivalence ratio, and the sensitivity to jet size decreases also.

Behavior in Relation to Primary Zones

From individual plots of overall NOx emissions against overall equivalence ratio for each jet size, values of NOx EI's at specific primary zone equivalence ratios can be extracted. From such values plots of overall NOx emissions as a function of jet size can be produced.

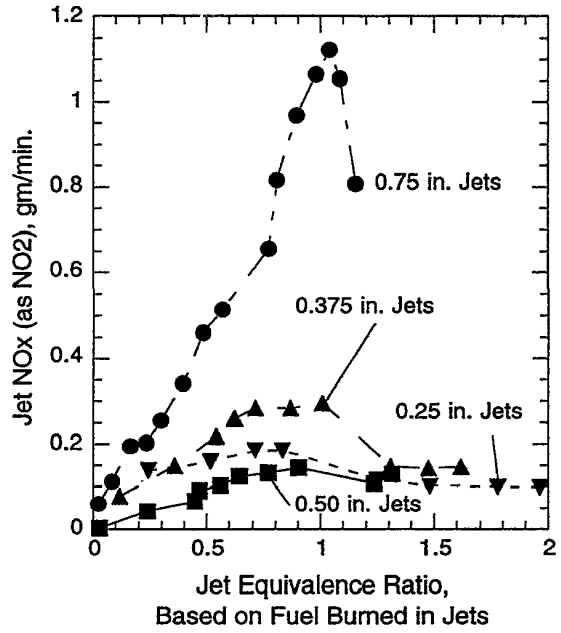

Figure 7: Absolute Values of Jet NOx

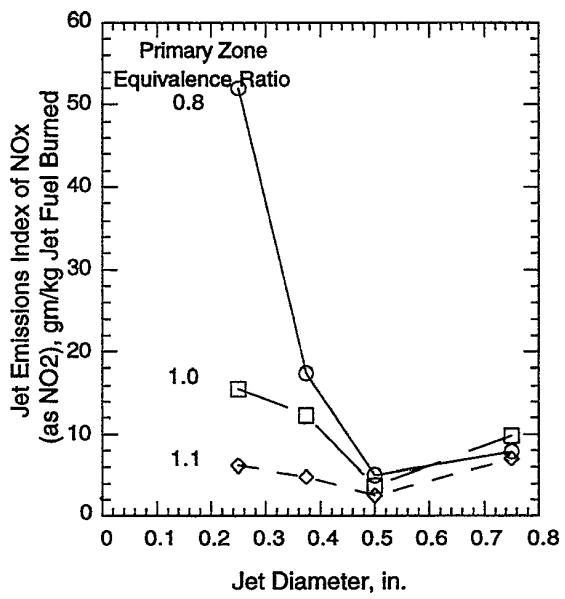

Figure 8: Indication of Optimum Jet Diameter

Figure 9 shows such a plot for primary zone equivalence ratios of 0.5, 0.8, 1.0 and 1.1. Since a particular primary zone equivalence ratio represents a specific bulk temperature, thermal NOx should be independent of jet size. That it is not, implies that the different jet sizes do result in some changes to flame

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structure in the jets, (since most of the NOx is generally generated in the jets) as was suggested, and that these changes are responsible for the observed increase in NOx.

Remembering that the overall NOx consists of NOx produced in the dome and NOx produced in the jets, the NOx EI for total fuel burned characteristics with jet diameter are similar at each primary zone equivalence ratio. NOx EI increases from the dome-alone value with the addition of combustion air jets. This is due to participation of some jet air in dome burning, as discussed above, and this increases with jet size for the 0.25 in. (6.35 mm) and 0.375 in. (9.53 mm) jets, together with a growing, and eventually dominating, contribution of jet NOx. At unity primary zone equivalence ratio and above, there appears to be an optimum jet size at 0.50 in (12.7 mm) diameter. There is a sharp increase in overall NOx EI at all primary zone equivalence ratios, that is associated with the use of 0.75 in. (19.05 mm) diameter jets. The lowest NOx at each of the primary zone equivalence ratios is achieved by the dome alone configuration.

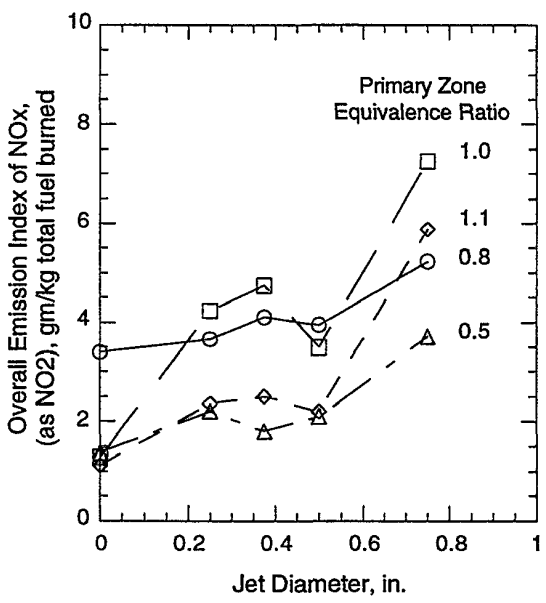

Figure 9: Optimum Jet Size for Overall Primary Zone

Figure 10 is a similar plot, but where the overall NOx is expressed as an absolute value, in gm/min., and is plotted against the mass flow rate ratio of jet air to dome air. In this plot at given primary zone equivalence ratio, the mass flow ratio varies as the jets are changed, and fuel flow rate increases with air mass flow ratio. Thus, as air mass flow rate ratio increases the dome gets progressively richer since its air flow is fixed.

The stoichiometry breakdowns for the combustor[2] show that for all but the highest air mass flow ratio shown on Figure 10, the dome is fuel-lean at 0.5 primary zone equivalence ratio, and is around stoichiometric or less at 0.8 primary zone equivalence ratio; the jet systems are always fuel-lean. Thus, primary zone equivalence ratios less than 0.8 generally represent overall lean burning in this combustor. For higher primary zone equivalence ratios, the dome is always very fuel-rich, and jet equivalence ratio becomes stoichiometric at unity primary zone equivalence ratio. Thus, primary zone equivalence ratios of unity or greater generally represent overall rich burning.

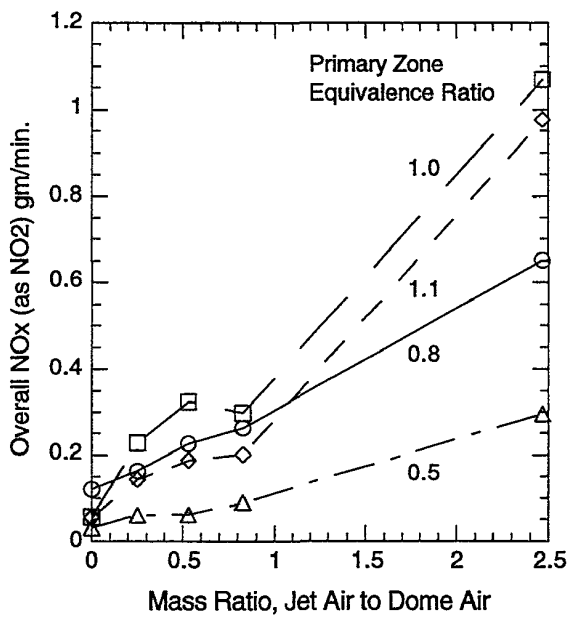

Figure10: Dependence of Absolute NOx Emissions on Jet to Dome Mass Flow Ratio

Figure 10 shows a clear difference in NOx characteristics between these two primary zone equivalence ratio regimes. For 0.5 and 0.8 primary zone equivalence ratios the absolute NOx level increases essentially linearly with increasing air mass flow rate ratio. As the primary zone equivalence ratio is increased the NOx levels increase due to the increased bulk temperature, and the sensitivity to air mass flow rate ratio is increased. At any primary zone equivalence ratio the increase in overall NOx with

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increase in air mass flow rate ratio is due to the growing contribution of jet NOx.

It is not sufficient to confine consideration to NOx alone as the indicator of low emissions; the inefficiency emissions must also taken into account.

Figure 11 is a plot of overall NOx emissions indices against overall CO emissions indices for each of the jet sizes, including the dome alone; the data for each jet size is limited on the left-hand-side by 0.5 primary zone equivalence ratio and on the right-hand-side by 1.1 primary zone equivalence ratio. Such a plot represents the emissions trades between NOx and CO that might be made. If a box is placed around the data for any particular jet size in the plot, bounded on one side by the maximum CO, on the top by maximum NOx, and, on the remaining side and the bottom by the respective axes, then that box is a qualitative measure of the total emissions production of the particular configuration. Data for any other configuration over the **same equivalence ratio range** that falls wholly ***inside*** the box, represents a **truly lower emissions configuration.**

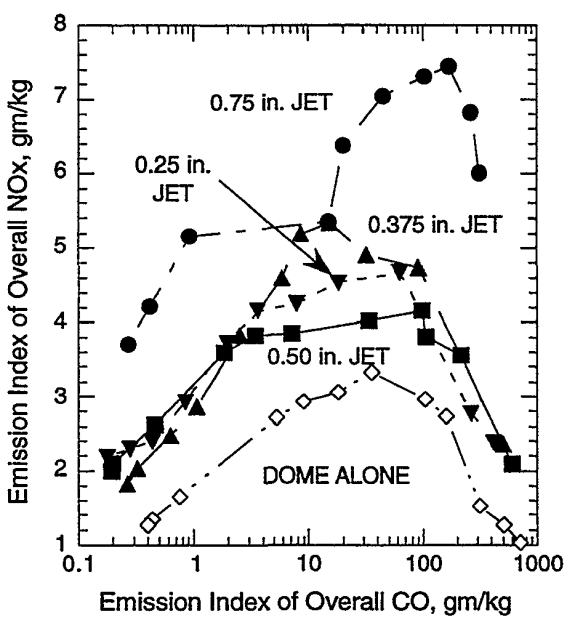

Figure 11: Overall Emissions Assessment for Range of Jet Sizes

Review of the figure reveals several interesting facts. First, the dome-alone configuration does truly represent the lowest emissions configuration, since it has the smallest box. Second, the 0.50 in. (12.70 mm) jet does indeed represent some kind of optimum jet configuration for this primary zone since its box is smaller than the boxes for the other jets, i.e. it does not achieve its low NOx by producing large amounts of CO. Third, the 0.75 in. (19.05 mm) jet is the worst emissions producer, and this is through its jet NOx generation.

The levels of CO over the equivalence ratio range in Figure 11 are all at or very close to, the appropriate equilibrium levels for the test conditions. For lower equivalence levels outside the range there is an upwards departure from equilibrium CO as a lean blowout condition is approached; similarly, at higher equivalence ratios there is also an upwards departure from equilibrium CO due to a lack of oxygen. Combustion efficiencies for the dome alone and for the 0.5 in. (12.70 mm) jet, for example, are essentially 100 percent to an overall equivalence ratio of about 0.8, and then fall off more or less together.

The finding that the dome alone configuration, i.e. no traverse air jets whatsoever, gives the lowest emissions, raises the valid question as to the need for the separate combustion air jets.

The combustion air jets are needed to reduce flame length and to enhance lean stability. Figure 12 shows some limited stability data for the dome alone and for the dome with the 0.50 in. (12.70 mm) jets, in the form of blowout overall equivalence ratio against the air loading parameter, ALP. The ALP is as defined in Reference 5. The blowouts were conducted at atmospheric pressure, with variation of dome pressure drop for the dome alone, and liner pressure drop at fixed dome pressure drop for the with-jet case. In addition for the dome alone case, two inlet air temperatures were used - ambient and 500F (533K). The plot shows that for given ALP the with-jets case blows out at lower equivalence ratio than does the dome alone. For the dome alone situation there were several flame modes observed[1]. Blowouts for the range of ALP's in Figure 12 were governed by lifted and attached flame conditions[6]; the presence of the combustion air jets exerts a back-pressure on the flame that modifies the flame behavior.

Figure 13 presents the dependency of overall equivalence ratio at blowout on jet size at fixed dome and liner pressure drops. It can be seen that the 0.50 in. jet size again represents an optimum for this dome.

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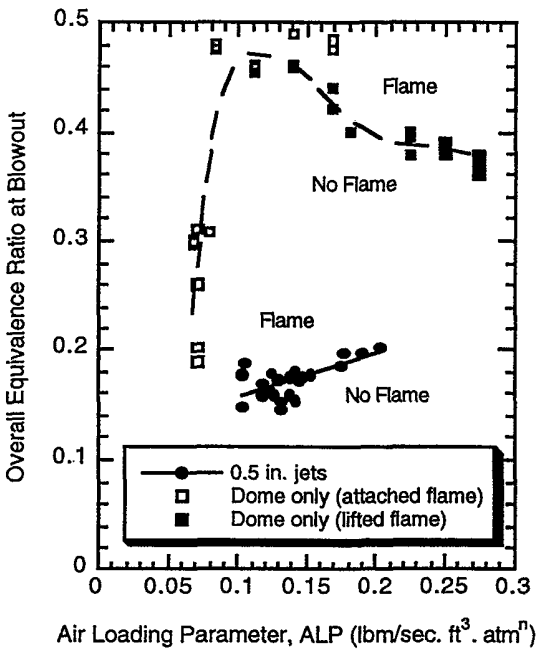

Figure 12: Stability Plot for Dome Alone and Dome with 0.5 in. Jets

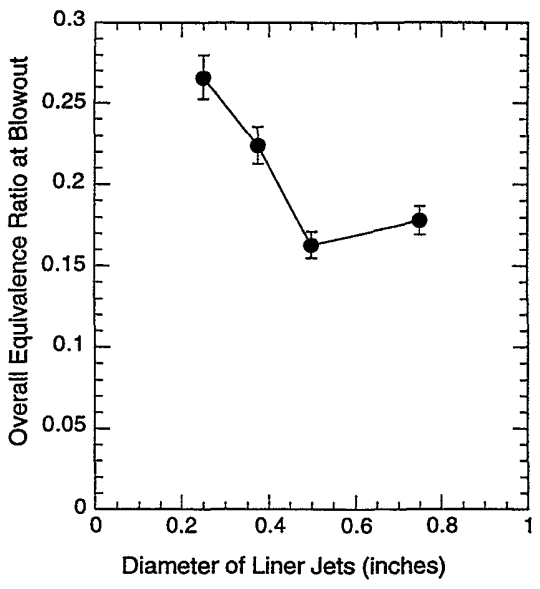

Figure 13: Blowout Equivalence Ratio as a Function of Jet Size

Discussion

Initial results, consisting of emissions measurements and lean blowout data, have been presented for a number of circular, single hole, opposed, combustion air jet sizes at a fixed liner position close to the dome, and at fixed dome and combustor liner pressure drops that are representative of engine design values. The absolute levels of emissions are dependent on the way that the simulations were conducted, and therefore the present results ideally should not be used for predictive purposes by scaling. However, the trends that are produced are valid, and will eventually (when complete) be useful for design purposes.

The series of emissions measurements gathered over wide ranges of test conditions for the different geometric configurations investigated, could form a useful data-set for evaluating the response of CFD modeling to realistic combustor geometries.

It was established that equivalence ratio was a suitable and unique parameter with which to consider the emissions measurements in this combustor and with the manner in which it was operated. The data have been analyzed in terms of a rich dome/lean jet burning behavior. This approach appeared to account nicely for the effects observed.

The combustor dome behaves as a partially stirred reactor, and shows significant unmixedness effects in its NOx characteristic. The partial set of results demonstrates that the jets play a significant role in generating the total NOx measured over equivalence ratio ranges that are typical of many primary zones. The jet contribution appears to arise from stoichiometric burning in the jets system. This stoichiometric jet burning is clearly identifiable for all but the smallest of the jets (0.125 in. dia.). As a consequence of the jet contributions to total NOx, the dome alone configuration emerges as the lowest producer of NOx and CO emissions. The jets are needed to shorten flame length, and to enhance lean blowout stability of the primary zone. Of the jet systems examined, that for the 0.5 in. (12.70 mm) diameter jet shows distinct optimum behavior in terms of minimum NOx emissions and maximum lean blowout stability.

The results to date indicate the need to use advanced optical diagnostic techniques to determine the exact flame structures existing in the jet systems, and how these are affected by the flow interactions between the dome and the jets. Detailed flow visualization is clearly most desirable. In particular, quantitative information on recirculation zone mass flow rates is essential. The parametric variations in liner pressure drop and combustion air jet axial

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position will also be of assistance in this respect. An explanation for the optimum behavior of the 0.50 in. (12.70 mm) jet must be obtained.

Although there was no attempt in the this combustor to achieve a true rich burn/quick quench/lean burn low emissions design, the present primary zone does function somewhat in this fashion. The findings here indicate how difficult it will be in such low emissions designs to achieve the quick quench portion and hence, to get a true lean burn.

The limited stability characteristics presented demonstrate that low emissions characteristics in a combustor cannot be pursued without regard to the other design criteria that must be satisfied for successful practical combustor designs.

Conclusions

1. Most of the NOx generated in the generic primary zone appears to originate from stoichiometric burning in the combustion air jet system.

2. Minimum total emissions are produced when all of the reactants are introduced through the combustor dome.

3. Combustion air jets are necessary to reduce flame length and confer lean blowout stability.

4. For at present undefined reasons, the 0.50 in. (12.70 mm) dia. opposed circular jets exhibit a clear optimum for the dome when the jets are positioned half a dome height downstream. This optimum is for minimum Nox, and lean blowout stability.

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References

1. Sturgess, G.J., *Combustor LBO Design Model Evaluation, Final Report,* (Draft) Pratt & Whitney Doc. No. FR-23400, 1 May, 1995

2. Gogeneni, S., Shouse, D., Frayne, C., Stutrud, J., and Sturgess, G., "Influence of Combustion Air Jets on Primary Zone Characteristics for Gas Turbine Combustors," proc. ISABE *12.th. Intl. Symp. on Air Breathing Engines,* editor F. Billig, Vol. 1, Melbourne, Australia, Sept. 10-15, 1995, pp. 475-487.

3. Roffe, G. and Venkataramani, K.S., "Emissions Measurements for Lean Premixed Propane/Air Systems at Pressures up to 30 Atmospheres," NASA Rept CR-159421, 1978.

4. Sturgess, G.J., McKinney, R.G. and Morford, S.A., "Modifications of Combustor Stoichiometry for Reduced Nox Emissions from Aircraft Engines," Trans. ASME, *J.Engrg. Gas Turbines an Power,* Vol. 115, No. 3, 1993, pp. 570-580.

5. Sturgess, G.J. and Shouse, D., "Lean Blowout in a Generic Gas Turbine Combustor with High Optical Access," ASME Paper No. 93-GT-332, May 1993, to appear in ASME Trans. *J. Engrg. for Gas Turbines and Power.*

6. Hedman, P.O., Sturgess, G.J., Warren, D.L., Goss, L.P. and Shouse, D.T., "Observations of Flame Behavior From a Practical Fuel Injector Using Gaseous Fuel in a Technology Combustor," Trans. ASME, *J. Engrg. for Gas Turbines and Power,* Vol. 117, No. 3, July 1995, pp. 441-452.